All results for: Neighbor Joining

Neighbor Joining Search All Forums

Your search returned approximately **15** results

Venkata R Gosula · 17 days ago Posted in Neighbor-Joining Algorithm

Found the following issues when solving Neighbor-Joining Algorithm:

1 - The order of the following 2 lines should probably be swapped:
remove *i*-th row and column as well as *j*-th row and column from *D*
add a new row/column *m* from *D* so that *Dk,m* = *Dm,k* = 1/2(*Dk,i* + *Dk,j* - *Di,j*) for any *k*
since the second line refers to recently deleted items.

2 - The complexity of trying to retain the original node numbers as in the original matrix, is more than expected, and not discussed at all in the algorithm.

3 - Extra Dataset has zeros merged with entries next to it on several lines, similar to what Erika Ramirez uncovered on the UPGMA extra dataset.

4 - Sample Output implies that floats should be rounded to 3 decimal places. But I did not pass the grader when I did that. When I allowed the full float value, only then I passed the grader. Note that the output for the extra dataset is not rounded to 3 decimal places either.

William Nicholson Signature Track · 17 days ago Posted in Neighbor-Joining Algorithm

The extra dataset for the neighbour-joining algorithm problem currently includes edge weights with negative values as well which is not consistent with the biological interpretation (i.e. the passage of time for mutations/ differences to arise), I think,

William

Phillip Compeau INSTRUCTOR · 14 days ago Posted in Another(not really) proof of Neighbor Joining Theorem

What you have proved is that if i has a neighbor j, then D*_i,j is the minimum element in the i-th row of D*.

Proving that D*_i,j is a minimum value of D* is more difficult.

Pavel Novikov · 14 days ago Posted in Another(not really) proof of Neighbor Joining Theorem

But if minimum element of each row corresponds to pair of neighbors it means that minimum element of all rows does too, and that's what the theorem asserts.

> **Neighbor-Joining Theorem:** Given an additive distance matrix *D*, a minimum element *D**_*i*,*j* of the neighbor-joining matrix *D** must correspond to neighboring leaves *i* and *j* in *Tree*(*D*).

Index i is arbitrary, so D*_i,j cannot always be minimum.
Or do you mean that assumption about each leaf's having neighbors is too strong?

Pavel Novikov · 14 days ago Posted in Another(not really) proof of Neighbor Joining Theorem

I've got another proof of the theorem and for me it seems a little beet more intuitive than the one in the lectures, but I'm not absolutely sure that I didn't miss anything. So if somebody could check it that would be very nice of them.

Assumptions and notations:
Let **T** be true tree. We'll assume that every leaf of $$T $$ has a neighbor and all internal edges have strictly positive lengths.
For pair of vertices $$a,b$$ let $$path(a,b)$$ denote set of edges that make up path between these vertices, and $$dist(a,b)$$ length of this path.
For any vertex $$a$$ let $$TotalDistance(a)$$ denote sum of distances from a to all the leaves(if $$a$$ is a leaf the definition agrees with the one from the lectures). Let $$Par(l)$$ - internal node next to $$l$$ for leaf $$l$$. So, $$D$$ is a distance matrix for leaves that correspond to $$T $$.

Proof:
Let $$i,j$$ be two arbitrary leaves of $$T$$.
So, $$D^{*}_{i,j} = (n - 2) \dot D_{i,j} - TotalDistance(i) - TotalDistance(j)$$.
$$TotalDistance(i)$$ is a sum of lengths of (n-1) paths. Each of these paths $$path(i,a)$$ consists of the edge $$(i,Par(i))$$ and $$path(Par(i), a)$$, so their total length = $$(n-1) \dot dist(i,Par(i))$$ plus sum of distances from $$Par(i)$$ to all leaves other than $$i$$. So, $$TotalDistance(i)$$ = $$TotalDistance(Par(i)) + (n-2) \dot dist(i,Par(i))$$

Next note, that $$D_{i,j} = dist(i, Par(i)) + dist(j, Par(j) + dist(Par(i), Par(j))$$. Thus, after contracting we get that
$$D^{*}_{i,j} = (n-2) \dot dist(Par(i), Par(j)) - TotalDistance(Par(i)) - TotalDistance(Par(j))$$ and first component is zero if $$i$$ and $$j$$ are neighbors. Without loss of generality let's assume that $$TotalDistance(Par(i)) \ge TotalDistance(Par(j))$$. Then it must be that if $$i$$ and $$j$$ are not neighbors already then we can get smaller element of $$D^{*}$$ by considering $$i$$ with its neighbor.

Erika Lorena Álvarez Ramírez · 7 days ago Posted in Small and Large Parsimony questions

As an answer to the third point, I applied Neighbor Joining to the Lafayette dataset, and then Large Parsimony, and LP improved the tree from 110 points to 109. Here it must not be difficult to invent a tree because there are 20 leafs, 19, I removed the repeated leaf, and is an unrooted binary tree, but with Neighbor Joining is easier to start. I haven´t checked what edges were exchanged but as it took

only one step, the tree didn´t change too much.
As a warning, in addition to the 20 aminoacids there is also the heavier X aminoacid

Sonia Keys · 8 days ago Posted in Stepic Additive Phylogeny | Step 4 does not display okay

much less neighbor-joining!

Erika Lorena Álvarez Ramírez · 16 days ago Posted in Neighbor-Joining Algorithm

I used this in the other problem, I´m just starting neighbor joining

Venkata R Gosula · 15 days ago Posted in Neighbor-Joining Algorithm

resolved.

Phillip Compeau INSTRUCTOR · 16 days ago Posted in Neighbor-Joining Algorithm

Please try solving this now as we have made some revisions and uploaded a new dataset.

Venkata R Gosula · 17 days ago Posted in Neighbor-Joining Algorithm

Good point. I did see the negative weights. But did not question their biological validity. What would they mean?

Pavel Novikov · 13 days ago Posted in Another(not really) proof of Neighbor Joining Theorem

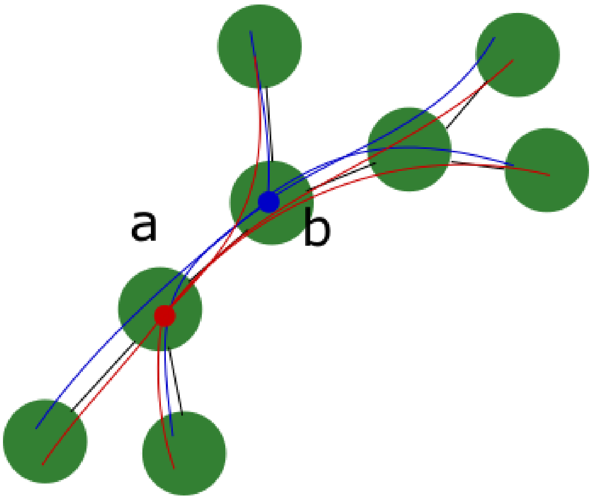
Upd: typos and an addition to consider the equality case.

As it was noted in the discussion that the proof above ignores important case of trees that contain leaves without neighbors. So that it wasn't left unfinished I try to add a completion. It is hardly easier than the one in the textbook, although.

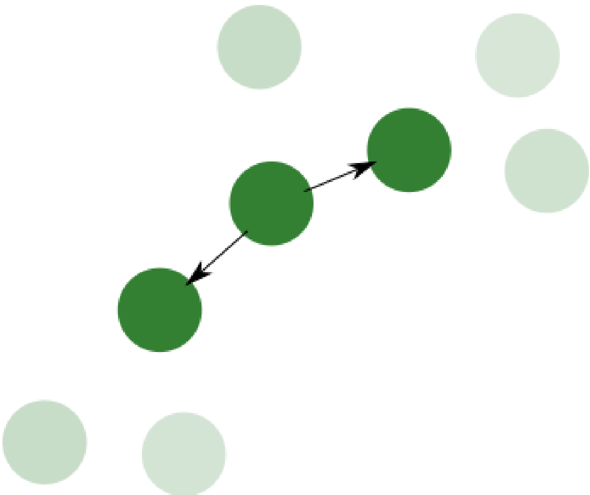
Assumptions and notations update:
Instead of assuming that every leaf has neighbors let's assume that all non-leaves have degree $$\ge$$ 3.
Let's call **protoleaf** any node that's connected to a leaf.
For any edge (a,b) $$Leaves_{a \rightarrow b}$$ will denote all the leaves that appear in the same component with $$a$$ if we cut this edge.
Proof update:
From the proof above it follows that if maximum of TotalDistance of all protoleaves corresponed to a protoleaf connected to more than one leaf, then the minimum element of D* would correspond to a pair of its neighboring leaves. What is left is to show that it can't correspond to a protoleaf with only one leaf.

Let's first compare TotalDistance of two arbitrary neighboring vertices $$a$$ and $$b$$ of the tree.

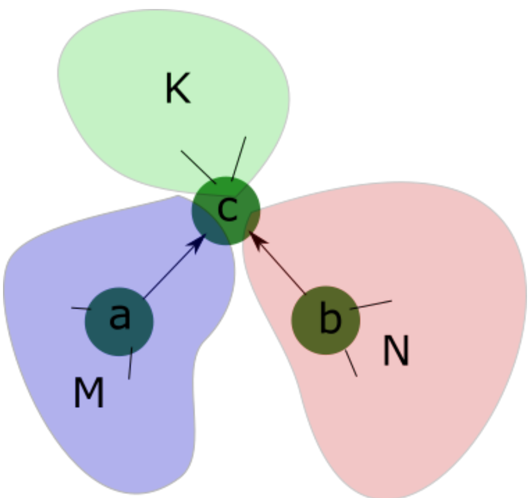
It's easy to see that $$TotalDistance(a) - TotalDistance(b) = dist(a,b) \dot (|Leaves_{b \rightarrow a}| - |Leaves_{a \rightarrow b}|)$$ and thus sign of this difference is determined only by numbers of leaves in subgraphs connected by this edge.



Let's now construct new oriented graph out of old one in the following way:
Leaves are left out. Each edge is directed towards vertex with bigger value of TotalDistance (that is, subgraph with smaller number of leaves). If values of TotalDistance are equal, there are two edges directed both sides, otherwise the edge will be called strict.



Let's assume there is a vertex $$c$$ with two incoming edges $$(a,c)$$ and $$(b,c) $$.
Denote $$M = |Leaves_{a \rightarrow c}|$$, $$N = |Leaves_{b \rightarrow c}|$$, $$K = |Leaves_{c \rightarrow b} \cap Leaves_{c \rightarrow a}|$$



Directions of arrow allow us to write inequalities $$K + N < M, K + M < N$$, which is only possible if $$K = 0$$ and that means that c has no incident edges other thatn (a, c) and (b,c) - and that contradicts our assumptions.

That means, that if a vertex in our oriented graph has degree > 1, it must have at least one strict outgoing edge.
And given the fact that all vertices in our directed graph, that are connected with less than two leaves in original graph have degree $$\ge$$ 2 we must be able to find a directed path that consists of strict edges from any protoleaf, connected to one leaf, to a protoleaf, connected to at least two, and that means that only there can be found the maximum of TotalDistance.

Venkata R Gosula · 15 days ago Posted in Neighbor-Joining Algorithm

I pass grader with latest dataset. Now rounding to 3 decimal places.

Verified that items 1,3 and 4 of my original post are resolved.

Item2 - it is what it is... not everything needs to be described in the algorithms.
Resolving Thread.

Erika Lorena Álvarez Ramírez · 8 days ago Posted in Small and Large Parsimony questions

Hi, I have some questions here:
-in the Small Parsimony problem, although the total score can be the same, the internal nodes can have different scores and allignments, or at least I don´t remeber if in the small dataset or in the extradataset I obtained different things, then, how reliable can be this method?

-now I have more doubts in the Large Parsimony problem, in the text there is a note saying that depending on how we break ties, we can obtain different scores at each step.I passed the grader but for the extradataset the minimum score there is 320 and as my program chose the way, the minimum score is 303, but I don´t know if this can be correct or there is something with my program. In fact, for the extradataset the score begins with 340 and then jumps to 335, and in total takes 9 steps to arrive to 320, and my program goes from 340 to 339 (it doesn´t choose the better 335) and takes 12 steps to arrive to 320 and then other 12 steps to arrive to a score of 303, could it be something with my program?, I tried to follow the instructions

-and last, what would be better, to invent a tree and apply Large Parsimony or to apply Neighbor Joining to know how the tree is, and then knowing the form of the tree, apply Small Parsimony?, or why Neighbor Joining is very used?, or they can give different trees because we lose information of the alignments if we don´t use parsimony, as the texts says?

thanks, sorry if I still don´t have the whole picture

Erika Lorena Álvarez Ramírez · 8 days ago Posted in Lafayette, repeated JT

Hi, is it correct that in the Lafayette dataset Janice´s strains be the same?, there are only 2 JT strains and in the figure there are 3, because there is one more patient in the dataset. With neighbor joining I don´t have problem because it only returns 2 edges with length 0, but for Parsimony I must delete one repeated leaf because of the way I wrote the program, thanks, ah, sorry if I wrote to much this day!, my apologies